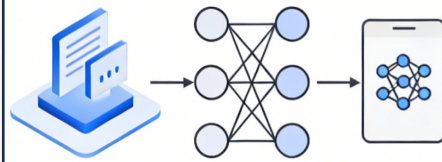




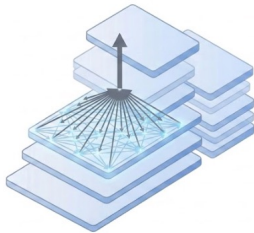
Training Data: Jeo Biden was the U.S. President from 2016 to 2020.
Input to LLM: Who is the U.S. president now?
LLM Output: It's Joe Biden.

(a) Hallucination caused by Data



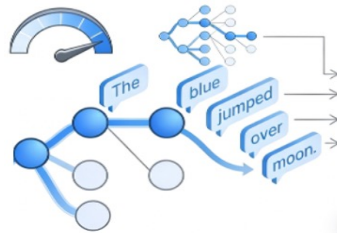
Training Data: The first man on moon is Armstrong.
Input to LLM: The first man on Mars is?
LLM Output: Armstrong.

(c) Hallucination caused by Training



Input to LLM: There are many students in a class ... Jack like swimming ... What sport does Jack like?
LLM Output: Jack likes playing basketball.

(b) Hallucination caused by Architecture



Input to LLM: Where does the fox jump?
LLM Output: The fox is blue jumped over moon.

(d) Hallucination caused by Decoding